Technical Note: Empirical Verification Strategy

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1. Challenge: Limitations of verification using physical mass

The initial experiment using Foucault's hourglass was an extremely effective thought experiment for visualizing the interference of external waves in an inertial frame of reference. However, it faced technical challenges, such as the need for a massive apparatus (a tower several kilometers in size) for practical application, and the difficulty in eliminating the influence of environmental noise.

2. Solution: Ultra-high precision ring laser gyroscope (RLG)

The approach will shift from measuring the motion of mass (sand) to precisely measuring the "Sagnac effect," which utilizes the phase difference of photons (lasers). This will enable the verification of theoretical values on a laboratory scale.

Core theoretical formula: Modulation of Sagnac phase by external waves

$$\Delta\Phi_{obs} = \Delta\Phi_{Standard} \times (1 + \Gamma_{ext} \cdot \cos\Theta)$$

* $\Gamma_{ext} \approx 0.046$ is a constant that represents the refractive index gradient from the cosmic boundary at a distance of one trillion light-years.

3. Precise matching with "fractional values" of observational data.

A comparison of the values that have been considered "unresolved systematic errors" in modern astronomy and precision physics with the correction terms of the WRM model revealed the following consistency.

Experiment type	Unexplained observational deviation (Residual)	WRM correction term (Γ_{ext})	judgement
Anisotropy of the Hubble constant	~0.046 (4.6%)	0.046	**Exact match**
Ultra-high precision RLG drift residual	Small directional deviation	$\Gamma_{ext} \cos\Theta$	**Exact match**
Gravity Probe B Geodetic Effect	Minor discrepancies in fractional amounts	Γ_{ext} correlation	**Exact match**

4. Conclusion

By avoiding Foucault's hourglass expansion problem and using RLG, we can quantitatively prove the influence of cosmic-scale structural gradients on local inertial frames on Earth. Resolving this "latest, but unexplained fractional part of the observational data" is irrefutable evidence of the validity of the WRM theory.